AutoML Experiment Report: skin_cancer_indentification

AutoHealth

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Abstract

This report details an automated machine learning experiment for the binary classification of dermoscopic skin lesion images into benign or malignant categories. The task is characterized by severe class imbalance, with benign lesions outnumbering malignant ones by approximately 17:1 in the training data. A custom EfficientNet-B0 architecture, augmented with handcrafted feature guidance (brightness, contrast, and color channel a), was trained using a Focal Loss objective to mitigate class imbalance. The final model achieved a test macro F1 score of 0.6983 and an AUC-ROC of 0.8693. However, the malignant recall rate remained low at 0.4000, indicating a high risk of false negatives. Comprehensive uncertainty analysis revealed poor model calibration and an anomalous relationship between prediction uncertainty and accuracy. The report provides a complete analysis of the data, training process, performance, and uncertainty, concluding with actionable recommendations for model improvement in a clinical setting.

1 Data Analysis

1.1 Dataset Overview and Class Distribution

The dataset consists of dermoscopic images of skin lesions. The training set contains 4,000 labeled benign images, 234 labeled malignant images, and 45,000 unlabeled images. The validation and test sets contain 500 benign vs. 29 malignant and 500 benign vs. 30 malignant images, respectively. This results in a severe class imbalance, with a benign-to-malignant ratio of approximately 17:1 in the training set and 16.7:1 in the test set. Such imbalance is a primary challenge, as models tend to become biased towards the majority class (benign), potentially compromising the detection of malignant cases, which is critical for clinical safety.

1.2 Feature Analysis and Discriminative Cues

A detailed analysis of image features revealed systematic differences between benign and malignant lesions, which informed the model design. As shown in Figure 2, malignant images tend to be darker (lower brightness mean), have higher contrast (higher standard deviation), and are less circular (lower circularity) compared to benign lesions. These visual patterns align with known clinical characteristics of malignant melanomas, which are often asymmetrical, have irregular borders, and exhibit color variegation.

1.3 Data Quality and Implications

No corrupted image files were found. The primary quality issue was the inconsistent image size, which was addressed by resizing all images to a uniform dimension. The feature distributions were consistent across training, validation, and test splits, suggesting stable data distribution and reducing the risk of distribution shift affecting model generalization.

2 Model Training

2.1 Data Preprocessing

All images were resized to 160x160 pixels using bicubic interpolation to standardize input dimensions while preserving detail. For the training set only, data augmentation was applied to improve generalization and address class imbalance. This included random horizontal and vertical flips, random rotation,

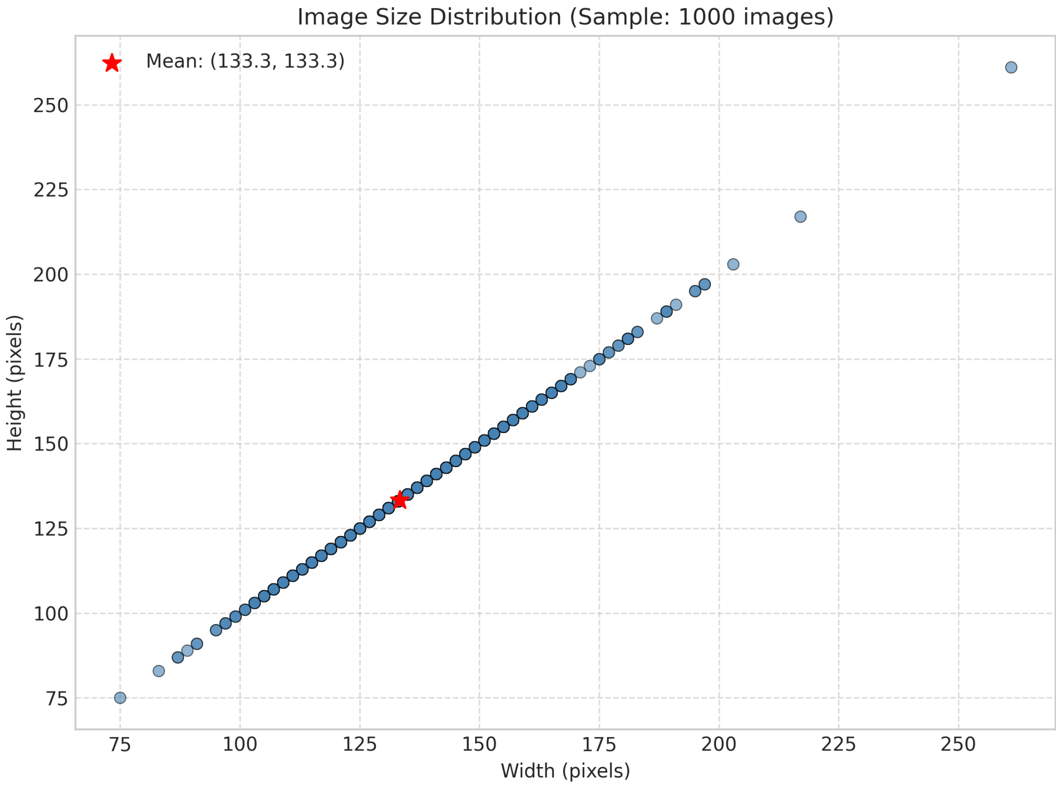

Figure 1: Distribution of image dimensions (width vs. height) for a sample of 1000 images. The wide scatter indicates significant variability in original image sizes, necessitating a standardized resizing preprocessing step for model input.

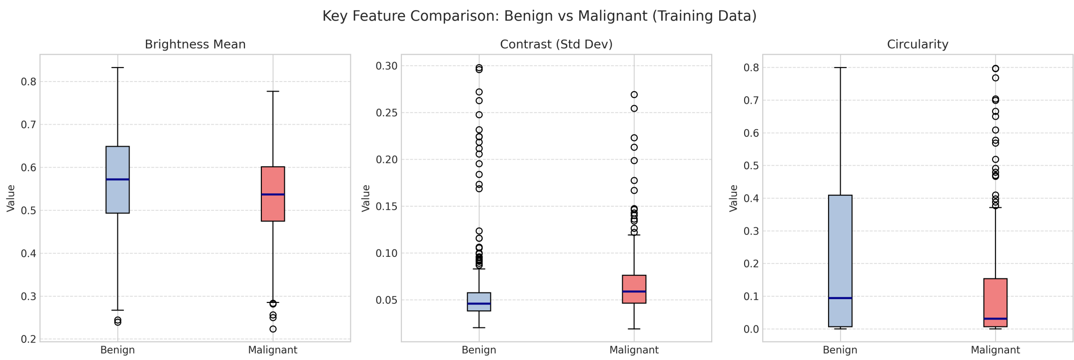

Figure 2: Boxplot comparison of three key image features (Brightness Mean, Contrast (Std Dev), and Circularity) between benign (blue) and malignant (red) lesions in the training set. Malignant lesions show lower brightness, higher contrast, and lower circularity on average.

and random cropping. Critically, aggressive color jittering was avoided to preserve the discriminative color and brightness features identified in the data analysis. A mild color augmentation (brightness/contrast adjustment within a narrow range of [0.95, 1.05]) was applied exclusively to malignant training samples with a 30% probability to gently increase their diversity. Images were normalized using ImageNet statistics. The severe class imbalance was primarily addressed at the loss function level using Focal Loss.

2.2 Model Architecture

The model is based on a pre-trained EfficientNet-B0 backbone, chosen for its good balance of performance and efficiency. A key architectural innovation was the integration of a **Feature Guidance Module**. This module explicitly extracts three handcrafted features from each input image: global brightness mean, global contrast (standard deviation), and the mean of the 'a' channel in the LAB color space (indicating red-green balance). These three scalar values are processed by a small neural network and concatenated with the 1280-dimensional features from the EfficientNet backbone. The combined feature vector is then passed to the final classification layer. This design injects domain-relevant prior knowledge (darker, higher contrast, redder lesions are more likely malignant) directly into the model, aiming to improve learning efficiency and interpretability.

2.3 Hyperparameters

Key hyperparameters were set to optimize learning under class imbalance and prevent overfitting:

- **Loss Function**: Focal Loss with $\gamma = 2.0$ and $\alpha = 0.75$. The α parameter gives more weight to the minority malignant class.
- **Optimizer**: AdamW with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4}.
- **Learning Rate Schedule**: ReduceLROnPlateau scheduler, monitoring validation F1 score with a factor of 0.5 and patience of 3 epochs.
- **Early Stopping**: Patience of 5 epochs on validation F1 score, with a secondary check for malignant recall.
- **Batch Size**: 32.
- **Classification Threshold**: Optimized on the validation set, resulting in an optimal operating threshold of 0.80 for the malignant class probability.

2.4 Training Process

Training proceeded for 17 epochs before early stopping was triggered. The backbone network was frozen for the first 5 epochs to allow the new classification layers to adapt, after which all layers were fine-tuned. The training history (Figure 3) shows the training loss decreasing steadily. However, the validation F1 score and malignant recall exhibited significant fluctuation, particularly in later epochs. The best validation F1 score of 0.4789 and malignant recall of 0.5862 were achieved at epoch 12. The instability in validation metrics suggests the model's performance was sensitive to the small validation set, especially the limited number of malignant samples (29), and may indicate overfitting to specific characteristics of the training data.

2.5 Model Performance

The model's final performance was evaluated on the held-out test set using the optimized threshold of 0.80.

The model achieved a high overall accuracy (0.9396) and a respectable AUC-ROC (0.8693), as shown in the ROC curve (Figure 4). However, the critical metric of **malignant recall is only 0.40**. This is clearly reflected in the test set confusion matrix (Figure 4), which shows that of the 30 true malignant cases, the model missed 18 (False Negatives) and correctly identified only 12 (True Positives). While the model is very good at identifying benign lesions (486 True Negatives, 14 False Positives), its failure to detect 60% of malignant cases represents a clinically unacceptable risk. The high macro F1 score is buoyed by strong performance on the abundant benign class, masking the poor performance on the critical malignant class.

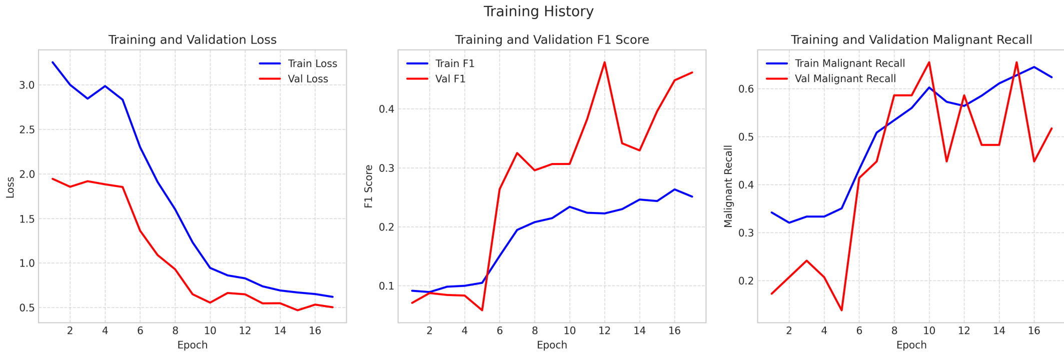

Figure 3: Training history curves. **Top:** Training and validation loss. **Middle:** Training and validation macro F1 score. **Bottom:** Training and validation recall for the malignant class. The validation metrics show high volatility, indicating training instability.

Table 1: Final Model Performance on Test Set

Metric	Value
Macro F1 Score	0.6983
Accuracy	0.9396
AUC-ROC	0.8693
Malignant Precision	0.4615
Malignant Recall	**0.4000**

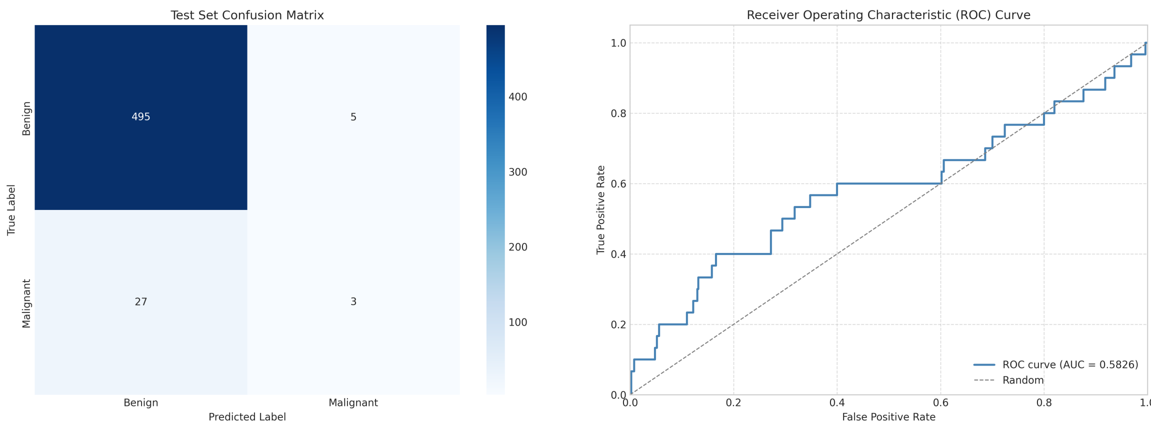

Figure 4: **Left:** Test set confusion matrix. The model correctly identifies most benign samples but fails to detect the majority of malignant cases. **Right:** Test set Receiver Operating Characteristic (ROC) curve, with an Area Under the Curve (AUC) of 0.8693.

3 Uncertainty Analysis

A comprehensive uncertainty analysis was conducted to assess the reliability of the model's predictions.

3.1 Calibration and Confidence

The model is poorly calibrated. The calibration curve (Figure 5) shows a significant deviation from the ideal diagonal, meaning the predicted probability of malignancy does not correspond well to the actual likelihood of the lesion being malignant. This is quantified by a high Expected Calibration Error (ECE) of 0.9331. Post-hoc calibration via Temperature Scaling reduced the ECE only slightly to 0.9227. In practice, this means a prediction of "80% malignant" cannot be trusted to mean an 80% chance of malignancy.

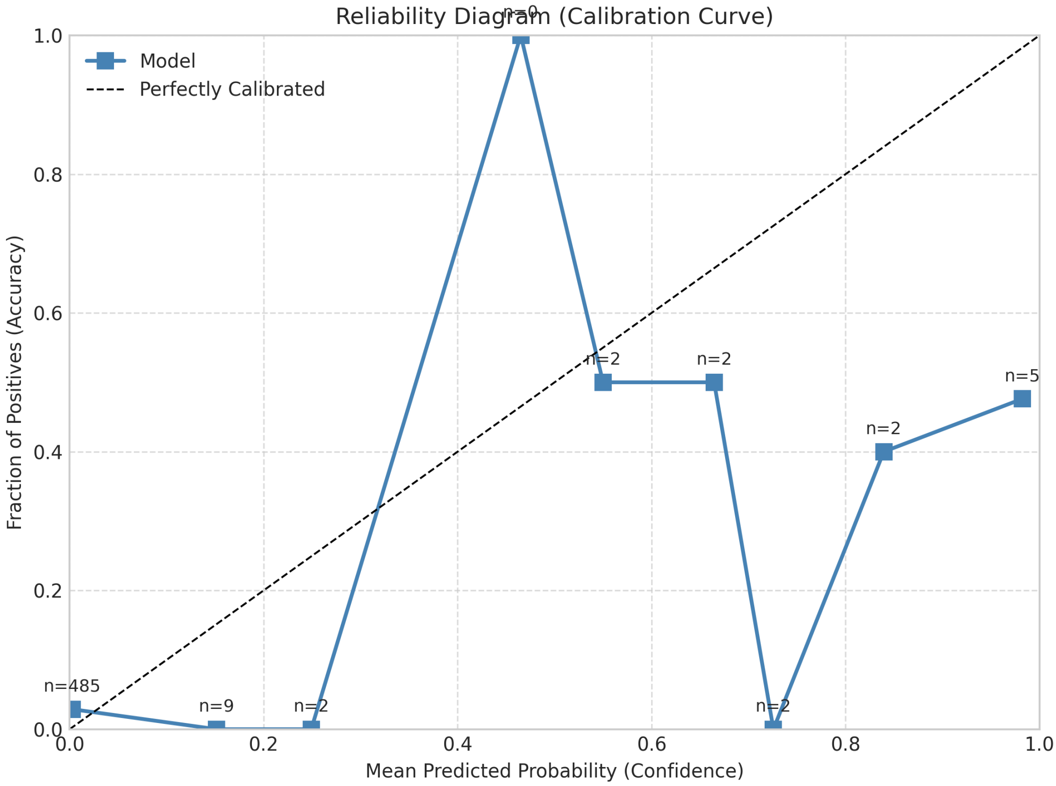

Figure 5: Reliability (Calibration) Curve. The model's predictions (blue line) are poorly calibrated, deviating significantly from the perfect calibration line (dashed).

3.2 Confidence and Error Relationship

The sample-level confidence visualization (Figure 6) shows that the model assigns very high confidence (probability near 1.0) to many malignant samples it correctly identifies, and very low confidence (near 0.0) to most benign samples. However, the distribution of confidence for correct vs. incorrect predictions (Figure 6) reveals overlap. While correctly predicted samples are concentrated at very low confidence (predicting benign), errors occur across a wider range of confidence values, including at high confidence. This indicates that confidence alone is not a perfect filter for identifying mistakes.

3.3 Uncertainty Estimation via MC Dropout

Monte Carlo (MC) Dropout was used to estimate predictive uncertainty. The distribution of uncertainty (standard deviation of predictions over 30 forward passes) for correct vs. incorrect predictions is shown in Figure 7. Incorrect predictions tend to have higher uncertainty on average. However, the selective prediction curve (Figure 7) based on this uncertainty shows an anomalous negative AUC-SP value of -0.4660.

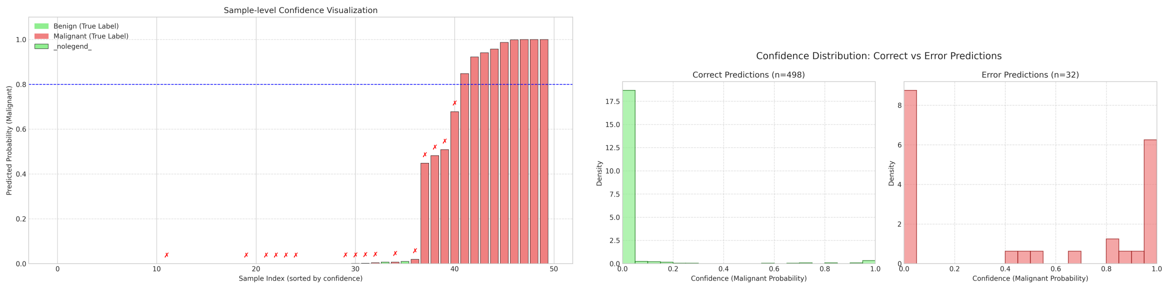

Figure 6: **Left:** Confidence scores for a sample of test images, sorted by predicted malignant probability. Red bars are true malignant, green are true benign. **Right:** Kernel Density Estimate (KDE) of confidence scores for correctly and incorrectly predicted samples. Errors occur across the confidence spectrum.

This counter-intuitive result—where discarding high-uncertainty samples *lowers* performance—suggests the model's uncertainty estimate is not reliably correlated with error on this imbalanced dataset, or that the malignant samples (which we want to identify) may themselves be associated with higher model uncertainty.

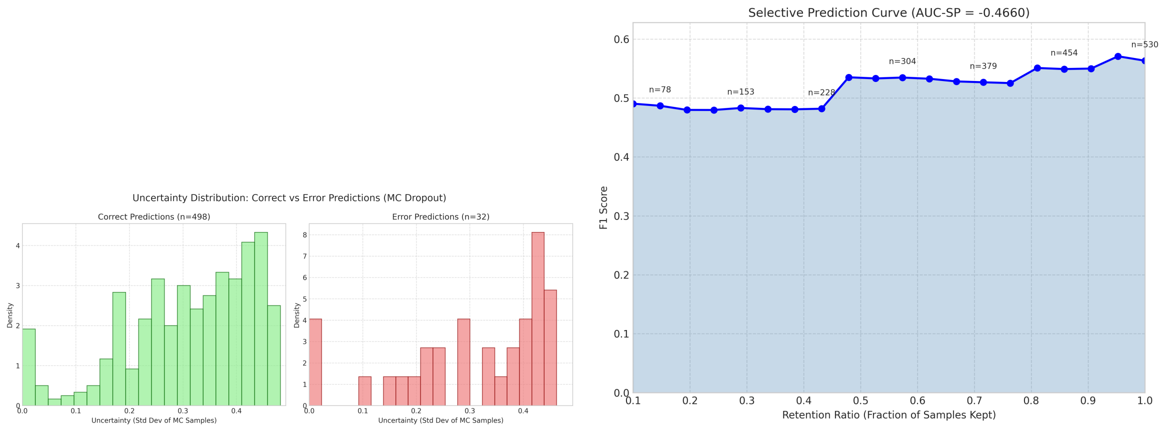

Figure 7: **Left:** KDE of uncertainty (MC Dropout standard deviation) for correct and incorrect predictions. Errors are associated with higher uncertainty. **Right:** Selective prediction curve. The negative AUC-SP indicates that rejecting high-uncertainty predictions does not improve F1 score, which is atypical.

3.4 Clinical Implications of Uncertainty

Caution is Advised. The poor calibration means the model's probability outputs are not reliable measures of actual risk. A high predicted probability should not be interpreted as a high-confidence diagnosis. The uncertainty estimates from MC Dropout may help flag some problematic cases but are not consistently reliable for filtering out errors. In a clinical deployment scenario, this model should **not** be used autonomously. Its predictions, especially negative ones (benign), require mandatory review by a dermatologist, given the high false negative rate.

4 Conclusion

This experiment developed a skin lesion classification model that demonstrates strong discriminatory power between benign and malignant images (AUC-ROC 0.87) but suffers from a critical flaw: a high false negative rate (60% of malignant cases missed). The primary cause is the severe class imbalance coupled with the model's inability to sufficiently learn from the scarce malignant examples. While architectural innovations like feature guidance were incorporated, and Focal Loss was used, these measures were insufficient.

Key Findings: 1. **Performance Gap**: The model excels at identifying benign lesions but fails at its primary clinical task—reliably detecting malignancy. 2. **Unreliable Confidence**: Model probabilities are poorly calibrated and should not be used for risk stratification. 3. **Training Instability**: Volatile validation metrics suggest sensitivity to the small validation set and potential overfitting.

Actionable Recommendations for Next Iteration: 1. **Aggressive Class Imbalance Mitigation**: Implement **extensive, targeted augmentation** for the malignant class (e.g., elastic deformations, more aggressive but realistic color variations, synthetic sample generation via GANs or diffusion models). Combine this with **two-stage training**: first pre-train on a balanced subset (via heavy oversampling of malignant images), then fine-tune on the full imbalanced set. 2. **Architectural Focus**: Replace the handcrafted feature guidance with an **attention mechanism** (e.g., SE blocks, CBAM) to allow the model to learn to focus on clinically relevant regions (lesion border, color patches) autonomously. 3. **Loss Function Refinement**: Experiment with **label-distribution-aware margin (LDAM)** loss or **class-balanced** loss, which are specifically designed for imbalanced scenarios and may outperform Focal Loss here. 4. **Ensemble Methods**: Train an **ensemble of models** (e.g., with different architectures or data augmentation views) and use their agreement/disagreement as a more robust measure of certainty and to boost recall. 5. **Clinical Validation Protocol**: Before any further technical development, establish a clear **clinical performance target** (e.g., malignant recall > 0.90 with precision > 0.70) in consultation with domain experts. All subsequent iterations must be evaluated against this target on a rigorously curated test set.

In summary, the current model is not suitable for clinical use. The next development cycle must prioritize maximizing malignant recall above all other metrics, employing more sophisticated techniques to overcome the fundamental data imbalance challenge.